Naïve Bayes Classifier

Naive Bayes Classifier

- Naive Bayes is a probabilistic machine learning algorithm that can be used in a wide variety of classification tasks.
- Bayesian classifier can **predict the probability** that a given tuple belongs to a particular class
- Naive Bayes classifiers are based on Bayes' Theorem.
- Bayesian classifier gives more speed and accuracy as compared to Decision tree classifier.

Bayes' theorem

- Bayes' theorem provides a way to calculate the probability of an event, based on prior knowledge related to that event.
- It's expressed as:

$$P(A|B) = P(B|A) * P(A) / P(B)$$

- P(A|B) is the **posterior probability** (Probability of hypothesis A on the observed event B.)
- P(B|A) is the **likelihood probability** (Probability of the evidence given that the probability of a hypothesis is true)
- P(A) and P(B) are the probabilities of events A and B.

Finding the probability of having a disease given you were tested positive
Finding the probability of liking Harry Potter given we know the person likes fiction

Why is it called ‘Naive’?

- The name naive is used because it assumes the features that go into the model is independent of each other.
- That is changing the value of one feature, does not directly influence or change the value of any of the other features used in the algorithm.

Naive Bayes Classifiers

- The dataset is divided into two parts, namely
- Feature matrix
- Response vector
- **Feature matrix** contains all the vectors (columns) of dataset in which each vector consists of the value of dependent features.
- **Response vector** contains the value of class variable for each row of feature matrix.

Working of Naive Bayes Classifiers

- Convert the given dataset into frequency tables.
- Generate Likelihood table by finding the probabilities of given features.
- Now, use Bayes theorem to calculate the posterior probability.

Naive Bayes Algorithm

Let the training set having n features

$F_1, F_2, F_3 \ldots F_n$ and feature vector

$\{f_1, f_2, f_3 \ldots f_n\}$

eg: {fast slow no.}

Let the class labels = $\{C_1, C_2, \ldots C_n\}$ and

Let the instance having feature vectors

$X = (x_1, x_2, x_3 \ldots x_n)$

Step 1 [Learning Phase] [Classification phase]

1. Compute the probabilities $P(C_k)$ for $k = 1, 2, \ldots P$.

Step 2.

Form a table showing conditional probability

$P(f_1|C_k), P(f_2, C_k) \ldots P(f_n|C_k)$

for all values of $f_1, f_2, f_3 \ldots f_n$ for $k = 1, 2, \ldots$

Step 3

Compute the products

$q_k = P(x_1|C_k)\, P(x_2|C_k) \ldots P(x_n|C_k)\, P(C_k)$

for $K = 1, 2, \ldots P$.

Step 4.

find j such $q_j = \max(q_1, q_2, \ldots q_p)$

(which class label we get max q_K)

Step 5

Assign the class label c_j to the instance X.

Example I

SL.No	Swim	Fly	Crawl	Class Label
1	Fast	No	No	fish
2	Fast	No	Yes	Animal
3	Slow	No	No	Animal
4	Fast	No	No	Animal
5	No	Short	No	Bird
6	No	Short	No	Bird
7	No	Rarely	No	Animal
8	Slow	No	Yes	Animal
9	Slow	No	No	Fish
10	Slow	No	Yes	Fish
11	No	Long	No	Bird
12	Fast	No	No	Bird

1. Construction of frequency table

Row → class labels

Column → features.

	Swim			Fly				Crawl		total
	fast	slow	no	long	long	Rarely	no	yes	no	
Animal	2	2	1	0	0	1	4	2	3	5
Bird	1	0	3	2	1	0	1	0	4	4
Fish	1	2	0	0	0	0	3	1	2	3

Step 1

Computation of probability of each class C_K

$K = 1, 2, 3$.

$$P(C_K) = \frac{\text{No. of samples with class label } C_K}{\text{total no. of samples.}}$$

$P(\text{Animal}) = 5/12.$

$P(\text{Bird}) = 4/12 = 1/3$

$P(\text{Fish}) = 3/12 = 1/4.$

Step 2: Construction of conditional probability table.

$$P(f_j | C_K) = \frac{P(f_j \wedge C_K)}{P(C_K)}$$

$$= \frac{\text{No. of Samples with } f_j \text{ \& } C_K}{\text{No. of samples with } C_K.}$$

Class	Swim			Fly		Rarely		Crawl	
	Fast	slow	No	short	Long		No	Yes	No
Animal	2/5	2/5	4/5	0/5	0/5	4/5	4/5	2/5	3/5
Bird	4/4	0/4	3/4	2/4	4/4	0	4/4	0/4	4/4
Fish	1/3	2/3	0/3	0/3	0/3	0/3	3/3	1/3	2/3

Step 3 Computation of conditional probability

$P(C_K|X)$, $K = 1, 2, 3$.

$X = (\text{slow}, \text{Rarely}, \text{No})$

$P(\text{Animal}|X) = P(\text{slow}|\text{Animal}) \cdot P(\text{Rarely}|\text{Anim}) \cdot P(\text{No}|\text{Animal}) \cdot P(\text{Animal})$

$= (2/5) \cdot (4/5) \cdot (3/5) \cdot (5/12)$

$= 0.02$

$P(\text{Bird} \mid x) = (0/4) \cdot (0/4) \cdot (4/4) \cdot (4/12)$

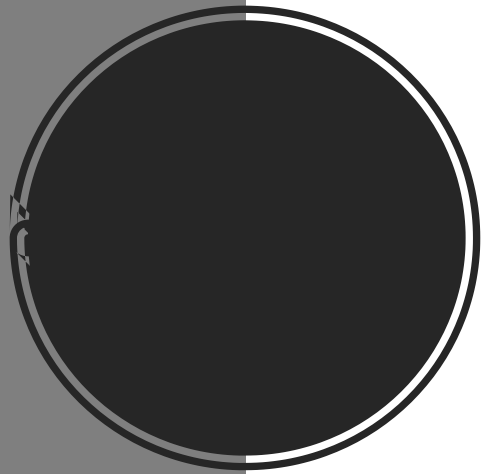
$= 0$

$P(\text{Fish} \mid x) = (2/3) \cdot (0/3) \cdot (2/3) \cdot (3/12)$

$= 0$

Step 4 Maximum probability is for $P(\text{Animal} \mid x)$

$\Rightarrow$ Animal $\rightarrow x \{$ slow, Rarely, No $\}$



Day	*Outlook*	*Temperature*	*Humidity*	*Wind*	*PlayTennis*
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Outlook = Sunny, Temperature = Cool, Humidity = High, Wind = Strong

	Outlook	Play
0	Rainy	Yes
1	Sunny	Yes
2	Overcast	Yes
3	Overcast	Yes
4	Sunny	No
5	Rainy	Yes
6	Sunny	Yes
7	Overcast	Yes
8	Rainy	No
9	Sunny	No
10	Sunny	Yes
11	Rainy	No
12	Overcast	Yes
13	Overcast	Yes

Advantages of Naive Bayes Classifiers

- Naïve Bayes is one of the fast and easy ML algorithms to predict a class of datasets.
- It can be used for Binary as well as Multi-class Classifications.
- It performs well in Multi-class predictions as compared to the other Algorithms.
- It is the most popular choice for text classification problems.

Disadvantages of Naive Bayes Classifiers

- Naive Bayes assumes that all features are independent or unrelated, so it cannot learn the relationship between features.

Applications of Naive Bayes Classifiers

- It is used for Credit Scoring.
- It is used in medical data classification.
- It can be used in real-time predictions because Naïve Bayes Classifier is an eager learner.
- It is used in Text classification such as Spam filtering and Sentiment analysis.